Model Immunization from a Condition Number Perspective

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Background: Foundation models and fine-tuning are transforming AI

- Open-weight encoders and vision-language models enable rapid adaptation to new tasks.
- Low-cost fine-tuning (linear probing, few-shot) drives wide deployment across domains.
- Optimization geometry (loss curvature) dictates adaptation speed and stability.
- Safety stakes: easy repurposing can enable harmful use with minimal effort.

Background: Why condition numbers matter for learning speed

- Gradient methods converge faster when the Hessian condition number κ is small.
- Ill-conditioned objectives slow, destabilize, or stall fine-tuning.
- Model features shape Hessians of downstream tasks; thus features control κ.
- Target: steer features so benign tasks are well-conditioned while harmful ones are ill-conditioned.

Problem: Selective model immunization

- Goal: make pre-trained models hard to fine-tune on harmful tasks, yet useful for benign tasks.
- Existing works show empirical effects but lack principled definitions and guarantees.
- Key challenge: selectively degrade optimization geometry only for harmful tasks.
- Need: theory clarifying when immunization is possible and how to enforce it.

Why is this hard?

- Benign and harmful tasks often share features; naive penalties hurt both.
- Optimization-based defenses must generalize to unknown future probes.
- No agreed metric to quantify "immunized" beyond final accuracy.
- Theoretical feasibility depends on data geometry across tasks.

Core idea and contributions

- They define and analyze immunization via Hessians for linear probing through a condition-number framework.
- They introduce a κ-maximizing regularizer to ill-condition harmful tasks.
- They present an end-to-end algorithm balancing κ increase (harmful) and κ decrease (pre-training).
- They report strong validation on linear models and deep nets with large RIR gains.

Framework: Linear probing and Hessians

- Setting: linear feature extractor θ; squared-loss linear probes on tasks.
- Task Hessian takes form $H(\theta) = \theta^\top K\theta$, where K is data covariance.
- $\kappa(H)$ controls gradient convergence speed for the probe.
- Therefore, shaping θ can reshape κ for downstream tasks.

Feasibility: Alignment between task covariances

- Singular values of $H(\theta)$ depend on alignment between θ and K's singular vectors.
- Selective immunization is feasible when K_{P} and K_{H} are sufficiently misaligned.
- Angles between principal directions dictate how κ can be steered.
- Takeaway: geometry of data covariances governs achievable selectivity.

Definition: What is an immunized model?

- Criteria: (a) $\kappa(H_{\mathrm{H}}(\theta_I))$ increases vs $\kappa(H_{\mathrm{H}}(I))$ on harmful tasks.
- (b) $\kappa(H_{\mathrm{P}}(\theta_I))$ does not increase vs $\kappa(H_{\mathrm{P}}(I))$ on pre-training.
- (c) Pre-training performance is maintained.
- Metric: Relative Immunization Ratio (RIR) $\gg 1$ indicates success.

Algorithm overview

- Objective: $\mathcal{R}_{\mathrm{ill}}(H_{\mathrm{H}}(\theta)) + \mathcal{R}_{\mathrm{well}}(H_{\mathrm{P}}(\theta)) + \mathcal{L}(\mathcal{D}_{\mathrm{P}}; \theta, \omega)$.
- Increase κ for harmful task; decrease κ for pre-training task; keep DP loss low.
- Closed-form gradients through $H(\theta) = \theta^\top K \theta$ enable end-to-end training.
- Covariance-normalized updates preserve κ monotonicity under conditions.

Method visual: κ predicts convergence (illustration)

- They illustrate how larger κ slows gradient convergence and smaller κ accelerates it.
- The plotted norm-ratio dynamics are consistent with the Hessian condition-number predictions.
- Takeaway: steering κ via θ translates into predictable optimization behavior.

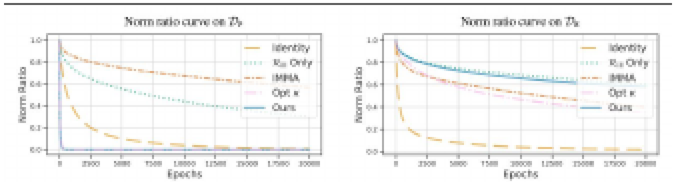

Figure: Norm ratio Eq. (16) vs. epochs under gradient descent with exact line search.

Component 1: κ-minimizing regularizer R_{well}

- Purpose: improve conditioning for the pre-training task.
- Property: monotonic κ decrease under gradient descent (from prior work).
- Effect: accelerates DP optimization and preserves utility.
- Used on $H_{\mathrm{P}}(\theta)$ to protect benign adaptation speed.

Component 2: New κ-maximizing regularizer $R_{\rm ill}$

- Form: $R_{\rm ill}(S) = \frac{1}{\frac{1}{2}k^{-1}\|S\|_F^2 - \frac{1}{2}(\sigma_{\min}(S))^2}$.
- Nonnegative; upper-bounds $1/\log\kappa$; differentiable when $\sigma_{\min}$ is unique.
- Gradient steps provably increase κ for suitable step sizes.
- Applied to $H_{\rm H}(\theta)$ to slow or destabilize harmful fine-tuning.

Training mechanics: gradients and practical details

- Closed-form ∇_θ for $H(\theta) = \theta^\top K\theta$ (both regularizers).
- Covariance-normalized updates; mild uniqueness/stepsize assumptions.
- Dummy layer for linearizing deep encoders (ResNet18, ViT).

Experimental setup: datasets, metrics, baselines

- Datasets: House Prices (regression), MNIST binary pairs (90), ImageNet pre-training vs Cars/Country211.
- Metrics: κ changes per task and Relative Immunization Ratio (RIR).
- Baselines: Rill Only (this work, ablation), IMMA (Zhou et al., NeurIPS 2023), Opt κ (this work, direct κ optimization), Init θ_0 (identity).
- Goal: achieve RIR $\gg 1$ while preserving DP accuracy.

Results: MNIST (90 binary tasks) — Table 2

- They report an average RIR ≈ 70.0 with success across all 90 pairs.
- Opt κ shows high variance and fails in many pairs; the method is consistent.
- The method preserves or improves benign conditioning while degrading harmful.
- Shows broad applicability across diverse task pairs.
- Baselines as in "Experimental setup"; IMMA: Zhou et al., NeurIPS 2023.

Table: Table 2. Quantitative results of immunization in MNIST (LeCun, 1998), computed over 3 random seeds and averaged over all digit pairs.

Method	**Eq. (15) (i)**$\uparrow$	**Eq. (15) (ii)**$\downarrow$	**RIR**$\uparrow$
Rill Only	14.832 ± 1.039	8.654 ± 0.606	1.933 ± 0.046
IMMA	4.522 ± 0.139	2.774 ± 0.094	1.774 ± 0.041
Opt κ	3.196 ± 1.225	0.756 ± 1.171	69.73 ± 54.00
Ours	6.345 ± 0.188	0.149 ± 0.009	70.04 ± 3.280

Results: Deep encoders (ImageNet → Cars, Country211) — Table 3

- The method achieves $\mathrm{RIR}_{\theta_0} \gg 1$ on both ResNet18 and ViT.
- ViT retains or improves ImageNet accuracy (utility preserved).
- ResNet18 shows modest DP accuracy impact with strong RIR gains.
- Confirms transfer of linear-theory insights to nonlinear deep nets.
- Baselines as in "Experimental setup"; IMMA: Zhou et al., NeurIPS 2023.

Table: Table 3. Quantitative results of immunization of model pre-trained on ImageNet (Deng et al., 2009), computed over 3 random seeds.

DH	Method	ResNet18				ViT			
		Eq. (17) (i)↑	Eq. (17) (ii)↓	RIR_{θ_0}↑	DP Test Acc. (%)↑	Eq. (17) (i)↑	Eq. (17) (ii)↓	RIR_{θ_0}↑	DP Test Acc. (%)↑
	Init. θ_0	1.0	1.0	1.0	68.24	1.0	1.0	1.0	81.78
Cars	R²I Only	1.878 ± 0.034	1.786 ± 0.025	1.057 ± 0.026	63.84 ± 0.292	13.121 ± 0.038	4.097 ± 0.098	3.342 ± 0.048	82.21 ± 0.035
	IMMA	0.866 ± 0.002	0.889 ± 0.001	0.974 ± 0.002	63.57 ± 0.234	1.422 ± 0.006	2.090 ± 0.043	0.702 ± 0.007	81.89 ± 0.010
	Opt κ	1.217 ± 0.021	0.798 ± 0.005	1.527 ± 0.019	63.65 ± 0.148	3.598 ± 0.510	0.171 ± 0.033	26.369 ± 2.814	82.51 ± 0.085
	Ours	2.386 ± 0.442	0.699 ± 0.062	3.467 ± 0.358	62.36 ± 0.173	7.945 ± 0.247	0.323 ± 0.086	34.517 ± 0.886	82.79 ± 0.200
Country211	R²I Only	20.727 ± 0.791	20.675 ± 1.685	1.038 ± 0.050	62.17 ± 1.599	69.291 ± 1.198	63.519 ± 6.620	1.122 ± 0.097	80.73 ± 0.129
	IMMA	0.791 ± 0.005	0.814 ± 0.006	0.972 ± 0.007	67.03 ± 0.146	6.242 ± 0.203	7.599 ± 0.717	0.845 ± 0.048	82.47 ± 0.036
	Opt κ	1.538 ± 0.155	1.053 ± 0.091	1.472 ± 0.043	66.81 ± 0.115	4.589 ± 0.079	0.300 ± 0.106	16.498 ± 5.183	82.79 ± 0.023
	Ours	3.287 ± 0.330	0.399 ± 0.034	8.714 ± 0.672	65.01 ± 0.143	20.894 ± 1.425	0.700 ± 0.082	41.341 ± 0.987	83.17 ± 0.075

Discussion: When is immunization possible?

- Feasibility hinges on alignment angles between K_{P} and K_{H}.
- Stronger misalignment ⇒ easier to increase DH κ while reducing DP κ.
- Limits: highly aligned tasks reduce achievable selectivity.
- In practice, the algorithm shows gains on deep models despite nonlinearity.

Limitations and practical considerations

- Uniqueness of extreme singular values is assumed for monotonicity proofs.
- Step-size tuning and covariance normalization are important in practice.
- Definition focuses on linear probes; nonlinear adapters may partially circumvent.
- Trade-offs: small DP accuracy drops are possible on some backbones.

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References (baselines)

- IMMA: Zhou, Y., Qin, C., Duan, R., Raghunathan, A., and Kolter, J. Z. IMMA: Immunizing Text-to-Image Diffusion Models Against Malicious Fine-Tuning. Advances in Neural Information Processing Systems (NeurIPS), 2023.

- Other baselines are internal to this work (Rill Only, Opt κ, Init θ_0).

Open Questions

- How can the theory extend to nonlinear extractors?
- What scheduling balances $\kappa(H_{\mathrm{P}})$ and $\kappa(H_{\mathrm{H}})$ automatically?
- Can multi-harm settings be handled without sacrificing DP utility?
- How does this interact with concept erasure and tamper-resistance?